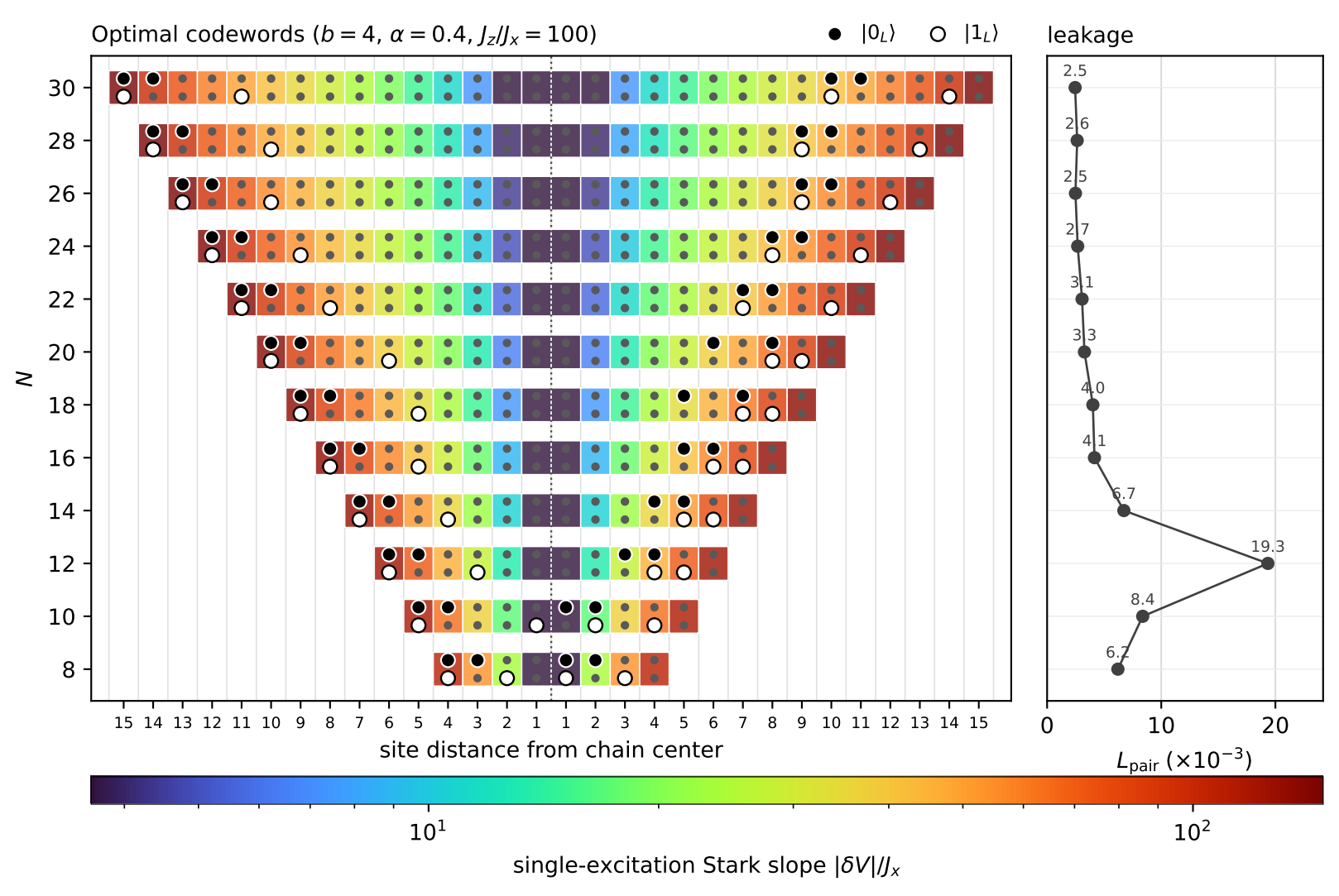

Optimal codewords ($b = 4$, $\alpha = 0.4$, $J_z/J_x = 100$)
$|0_L\rangle$
$|1_L\rangle$
leakage
N
30
28
26
24
22
20
18
16
14
12
10
8
15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
site distance from chain center
2.5
2.6
2.5
2.7
3.1
3.3
4.0
4.1
6.7
19.3
8.4
6.7
0
10
20
L_{pair} ($\times 10^{-3}$)
10^1
10^2
single-excitation Stark slope $|\delta V|/J_x$